

Rewritten Query $\tilde{q}$
async_fetch_query_embedding
// BM25 (no embedding)

BM25
fields: issue², issueDesc
fetch=150 (THREE_WAY_BM25_FETCH)

KNN text_vector
issue embedding
fetch=10, k=100

KNN desc_vector
issueDesc embedding
fetch=10, k=100

merge_rrf_hits (three_way)
K=40 (RRF_RANK_CONSTANT)
w_bm25=1.05, w_text=0.85, w_desc=1.0
top_n=50 (CS_KB_RRF_TOP_N)

Reranker (CS_KB_RERANK_ENABLED)
input=50 $\rightarrow$ top_k=10 (cs_kb_top_k)
timeout=5s, max_tokens=500

RRF fallback
incomplete config / API failure
truncate top_k by RRF order

Ranked Hits C_faq
top_k=10 $\rightarrow$ LLM candidate pool

success

